



IBM Developer  
SKILLS NETWORK

# Winning Space Race with Data Science

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# Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix



# Executive Summary

- A collection of data from public SpaceX API and Space X Wikipedia were compiled. Created a column 'Class' which represents success landings. Examined data using SQL, visualization, Folium maps, and dashboards. Collected relevant columns/variables to be used as features. Altered all categorical variables to binary using one hot encoding, standardized the data, and used GridSearch CV to find the best parameters for machine learning models. Accuracy score of all models put into a visualization.
- Logistic Regression, Support Vector Machine, Decision Tree Classifier, and K Nearest Neighbors are the four machine learning models used. All formulated similar results with accuracy scores at 83.33%. All models over predicted successful landings, therefore more data is needed for better model accuracy.



# Introduction



## BackGround:

- Growing industry of Commercial Space
- Space X with the best pricing (\$62 Million vs. \$165 Million USD)
- Salvageable 'Stage 1' part of rocket
- Space Y wants to be a competitor of Space X

## Problem:

- Training a Machine Learning Model to predict successful Stage 1 recovery





Section 1

# Methodology

# Methodology

## Executive Summary

- Data collection methodology:
  - Combined data from SpaceX public API and SpaceX Wikipedia page
- Perform data wrangling
  - Classifying 'Successful' true landings and 'Unsuccessful' false landings
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Models tuned using GridSearchCV

# Data Collection

The collection process of data involved API requests from SpaceX public API and web scraping data from a table in the SpaceX's Wikipedia page.

The following slide will show a flowchart of data collection from API and the slide after next will show a flowchart from web scraping.

## Space X API Data Columns:

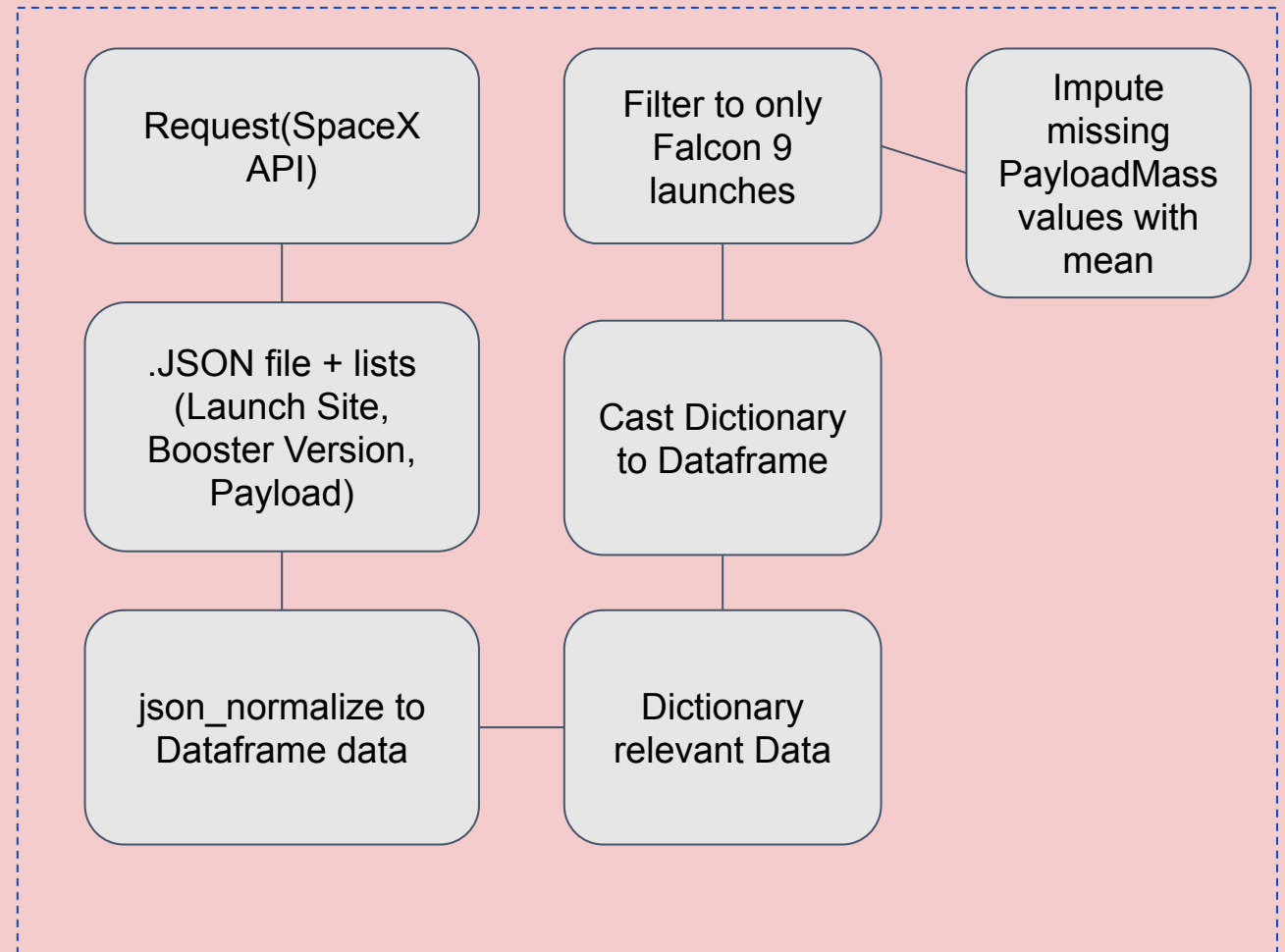
- Flight number, Date, BoosterVersion, PayloadMass, Orbit, LaunchSite, Outcome, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount, Serial, Longitude, Latitude

## Wikipedia Webscrape Data Columns:

- Flight No., Launch site, Payload, PayloadMass, Orbit, Customer, Launch outcome, Version Booster, Booster landing, Date, Time

# Data Collection – SpaceX API

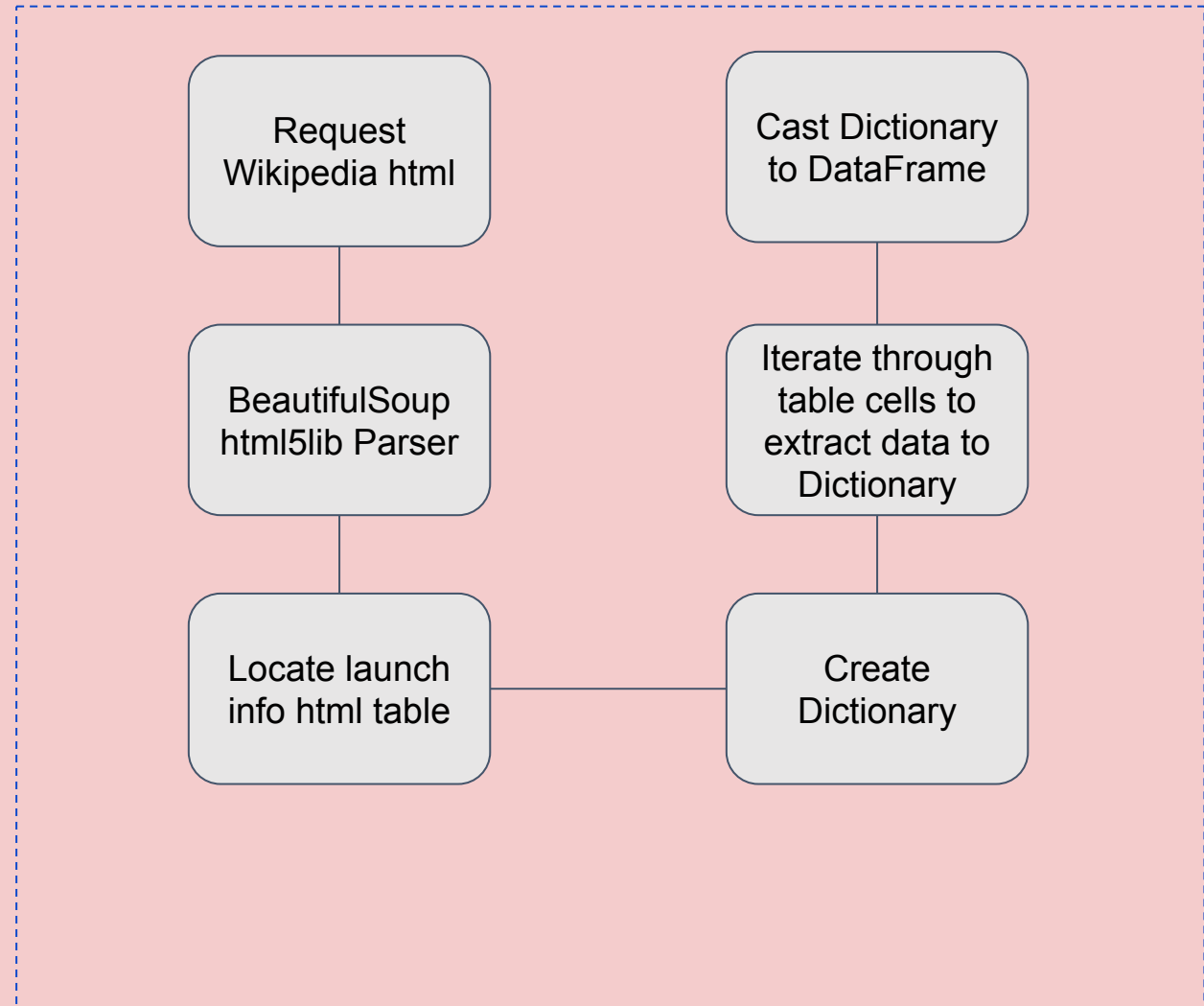
- **Data Collection = SpaceX API**
- <https://github.com/Blasko1369/SpaceY/blob/main/jupyter-labs-spacex-data-collection-api.ipynb>





# Data Collection - Scraping

- **Data-Collection = Web Scraping**
- <https://github.com/Blasko1369/SpaceY/blob/main/jupyter-labs-webscraping.ipynb>



# Data Wrangling

- Training Label was created to show landing outcomes where successful = '1' and unsuccessful = '0'.
- Outcome column includes 2 components: 'Mission Outcome' and 'Landing Location'
- A new training label column 'Class' will have a value of 1 if 'Mission Outcome' is True and a value of 0 if False.

## Value Mapping:

True ASDS, True RTLS, and True Ocean - set to => 1

None None, False ASDS, None ASDS, False Ocean, False RTLS - set to => 0

GithubURL:

<https://github.com/Blasko1369/SpaceY/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>

# EDA with Data Visualization

- Exploratory Data Analysis performed on variables Flight Number, Payload Mass, Launch Site, Orbit, Class and Year

## Plots Used:

- Flight Number vs Payload Mass, Flight Number vs Launch Site, Payload Mass vs Launch Site, Orbit vs Success Rate, Flight Number vs Orbit, Payload vs Orbit, and Success Yearly Trend
- Scatter Plots, Line Charts, and Bar Plots were used to compare relationships between variables to decide if a relationship exists to determine if they can be used in training the machine learning model.
- GitHub URL: <https://github.com/Blasko1369/SpaceY/blob/main/edadataviz.ipynb>



# EDA with SQL

- Data Loaded into SQLite database
- Queried using SQL Python integration
- Queries made to better understand dataset
- Queried information about launch site names, mission outcomes, various payload sizes of customers and booster versions, and landing outcomes

Github URL:

<https://github.com/Blasko1369/SpaceY/blob/main/jupyter-labs-eda-in-sql>

# Interactive Map with Folium

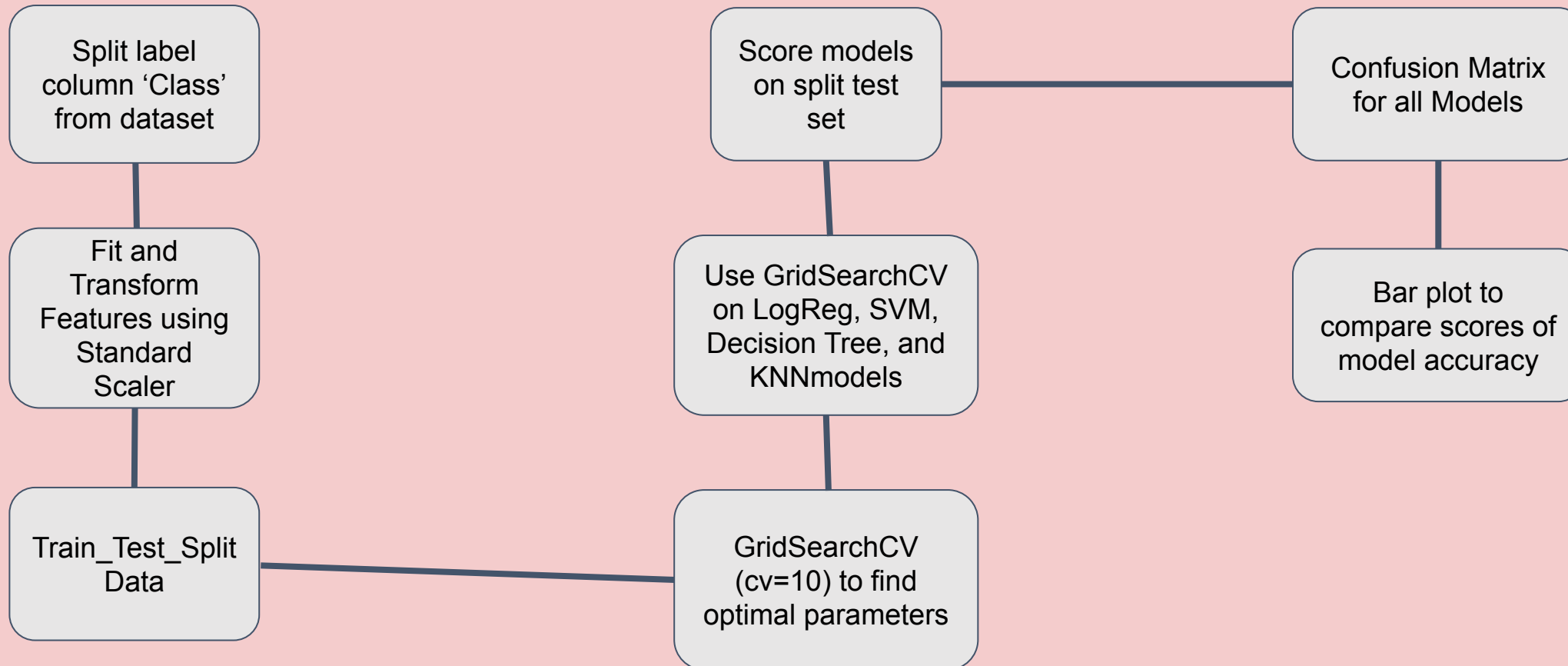
- Launch Sites, 'Successful' and 'Unsuccessful' Landings, and a proximity example to key locations: Railway, Highway, Coast, and City
- This allows us to see why launch sites are located where they are and also helps visualize successful landings in relation to launch location.
- GitHub URL: [https://github.com/Blasko1369/SpaceY/blob/main/lab\\_jupyter\\_launch\\_site\\_location.ipynb](https://github.com/Blasko1369/SpaceY/blob/main/lab_jupyter_launch_site_location.ipynb)

# Dashboard with Plotly Dash

Two diagrams were used to show findings:

- **Pie Chart** is used to visualize launch site success rate.
  - The **Pie Chart** can be selected to show distribution of successful landings across all launch sites and selected to show individual launch sites.
  - **Scatter Plot** takes two inputs: All sites/Individual site and Payload Mass on a slider between 0kg and 10000kg.
  - **Scatter Plot** helps us see how successful landings varies across launch sites, payload mass, and booster version category.
- 
- GitHub URL: <https://github.com/Blasko1369/SpaceY/blob/main/spacex-dash-app.py>

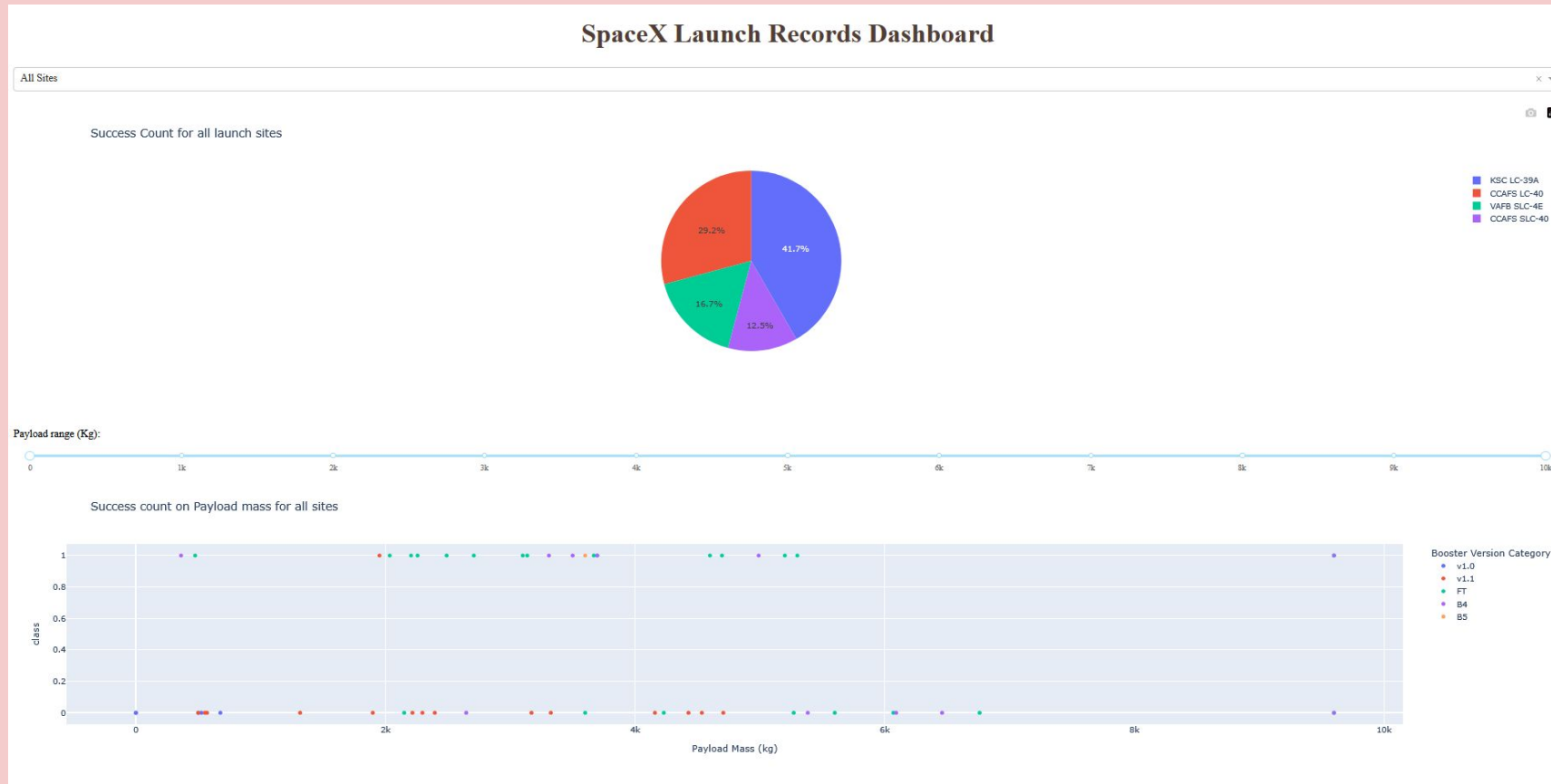
# Predictive Analysis (Classification)



- GitHub URL:  
[https://github.com/Blasko1369/SpaceY/blob/main/SpaceX\\_Machine%20Learning%20Prediction\\_Part\\_5.ipynb](https://github.com/Blasko1369/SpaceY/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb)

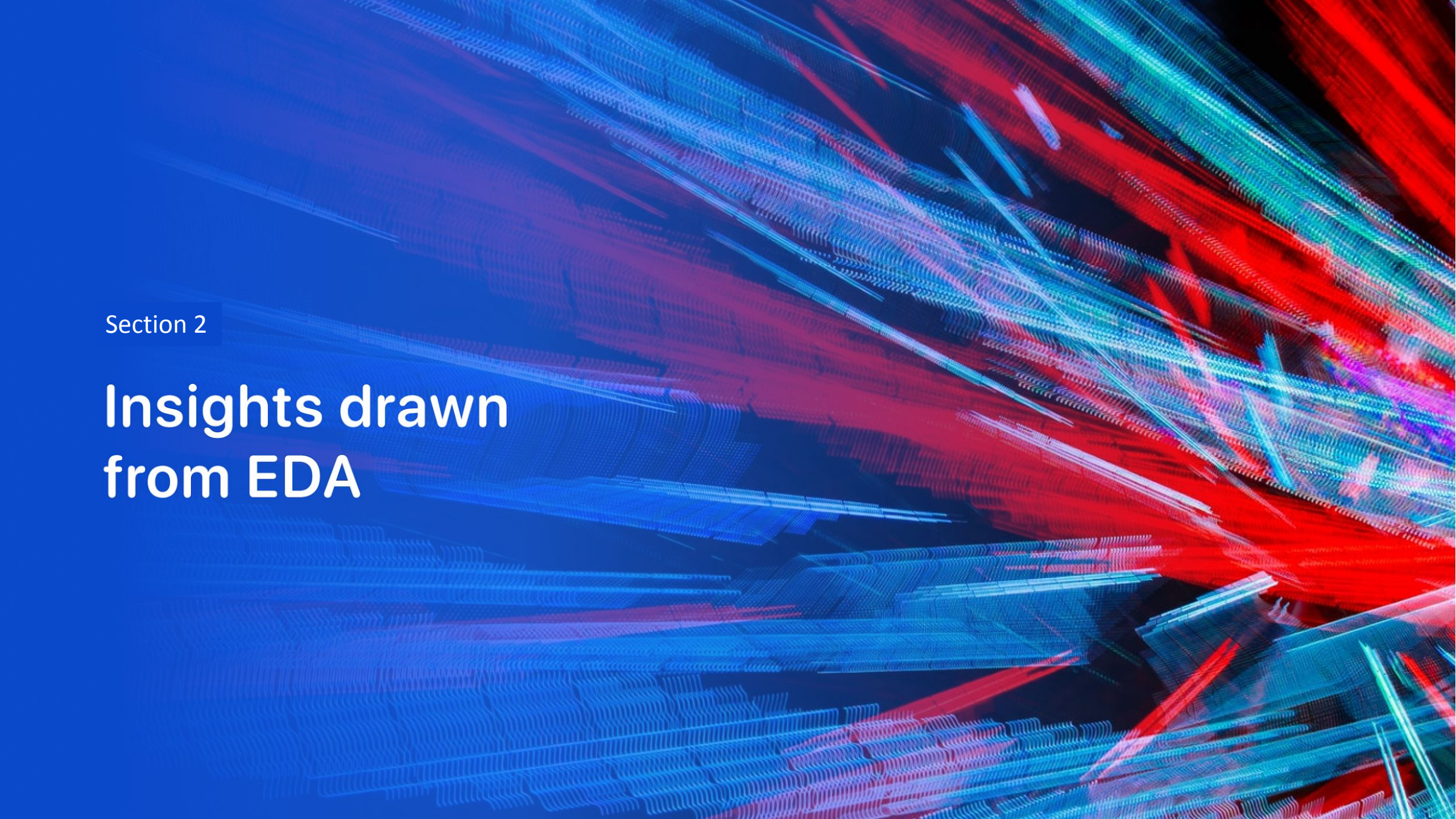


# Results



This is a preview of the Plotly Dashboard. The following slides will show the results of EDA with Visualization, EDA with SQL, Interactive Map with Folium, and the results of the model with 83.33% accuracy.



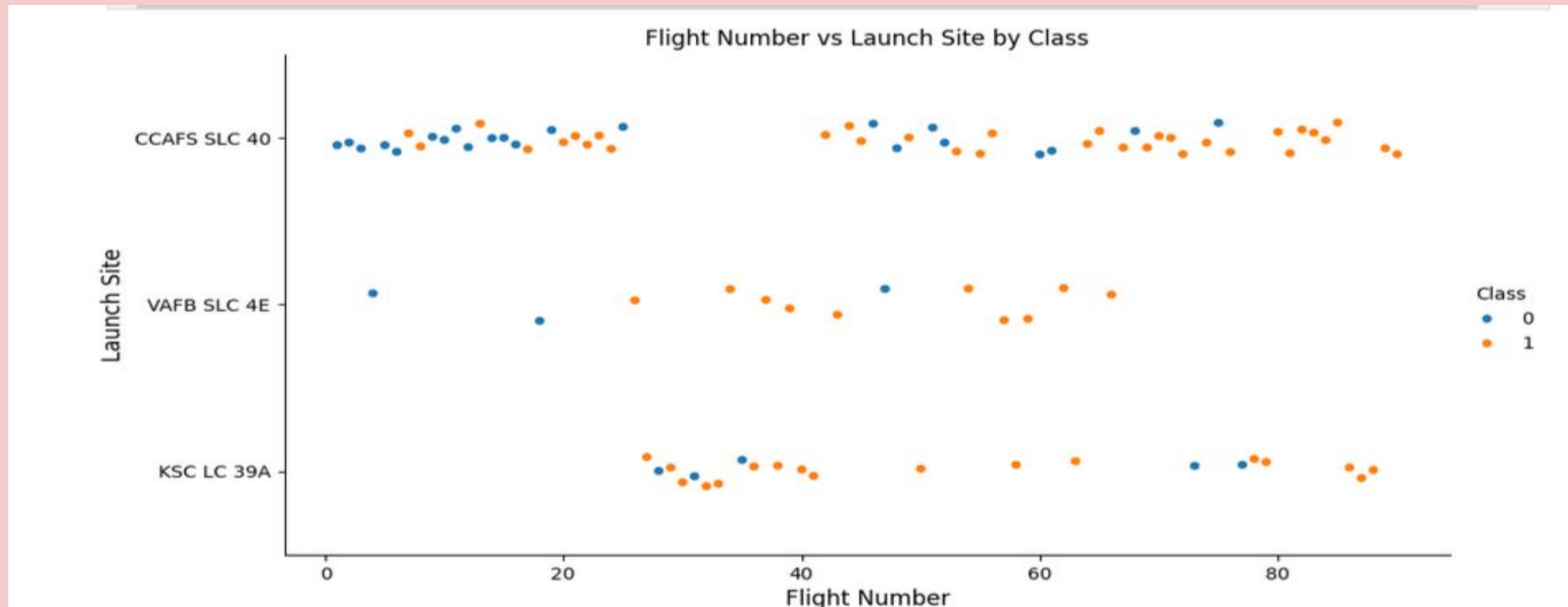
The background of the slide is an abstract composition. It features a solid blue area on the left side, which transitions into a complex pattern of diagonal streaks in shades of blue and red on the right. These streaks are layered over a fine, light-colored grid, creating a sense of depth and digital complexity.

Section 2

# Insights drawn from EDA



# Flight Number vs. Launch Site



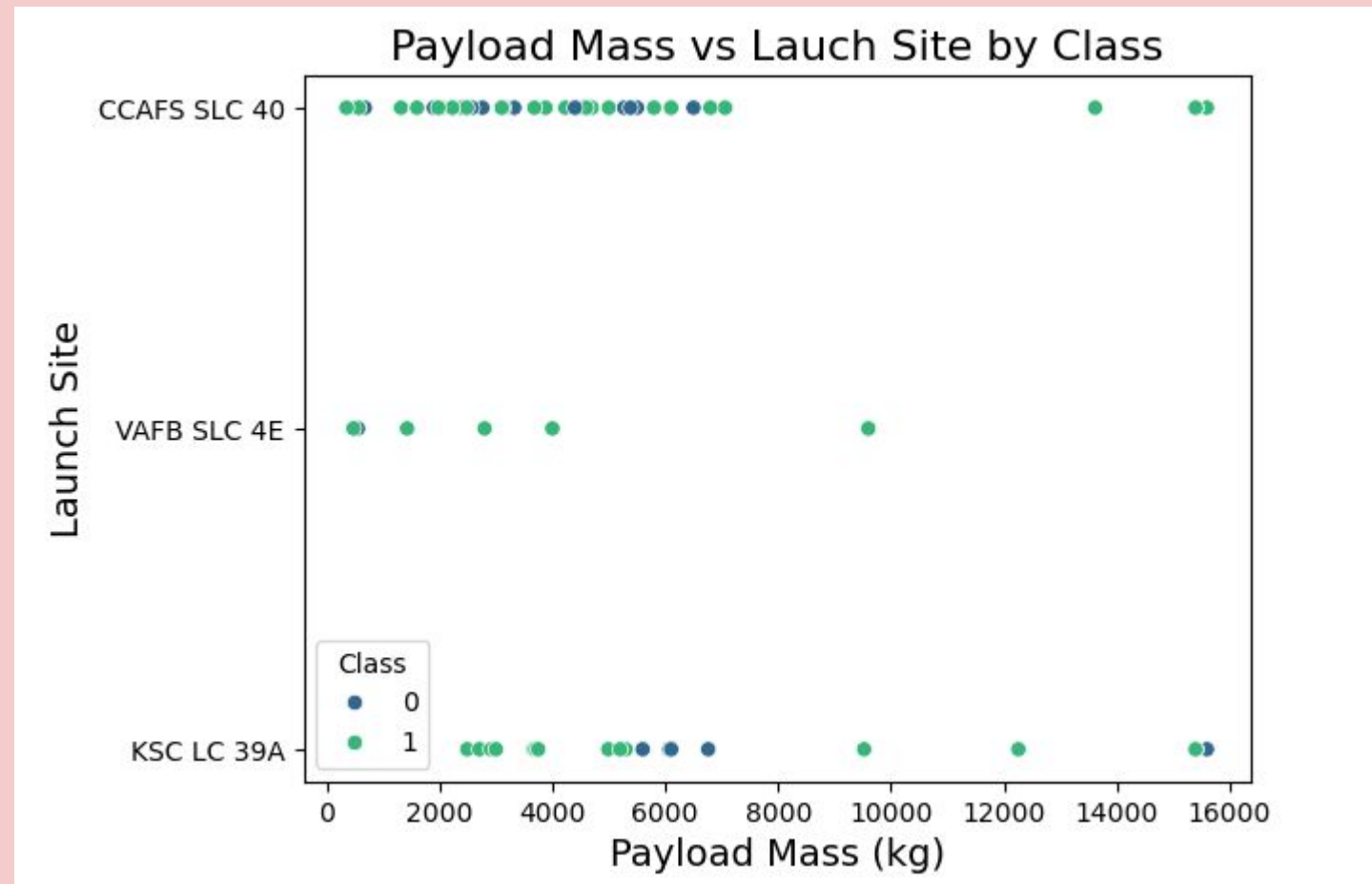
Orange = Success  
Blue = Unsuccessful

The Scatter Plot suggests an increase in success rate over time (indicated in Flight Number). Around the 20th flight we notice a significant increase in success rate. CCAFS appears to be the main launch site, for it has the most volume.

# Payload vs. Launch Site

Payload mass appears to fall mostly between 0kg - 6000kg.

Different launch sites also seem to use different payload mass.



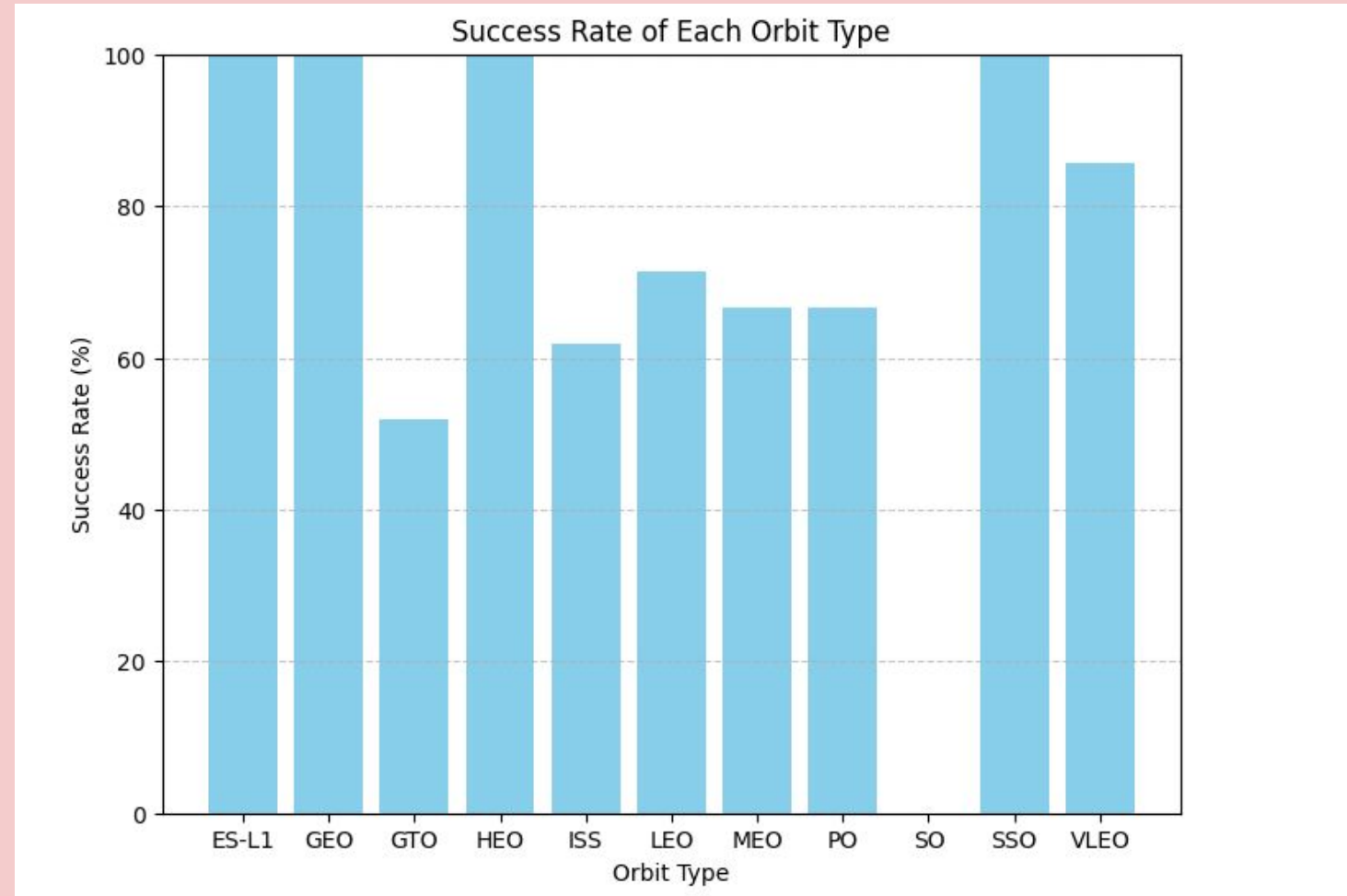
Green = Successful

Blue = Unsuccessful

# Success Rate vs. Orbit Type

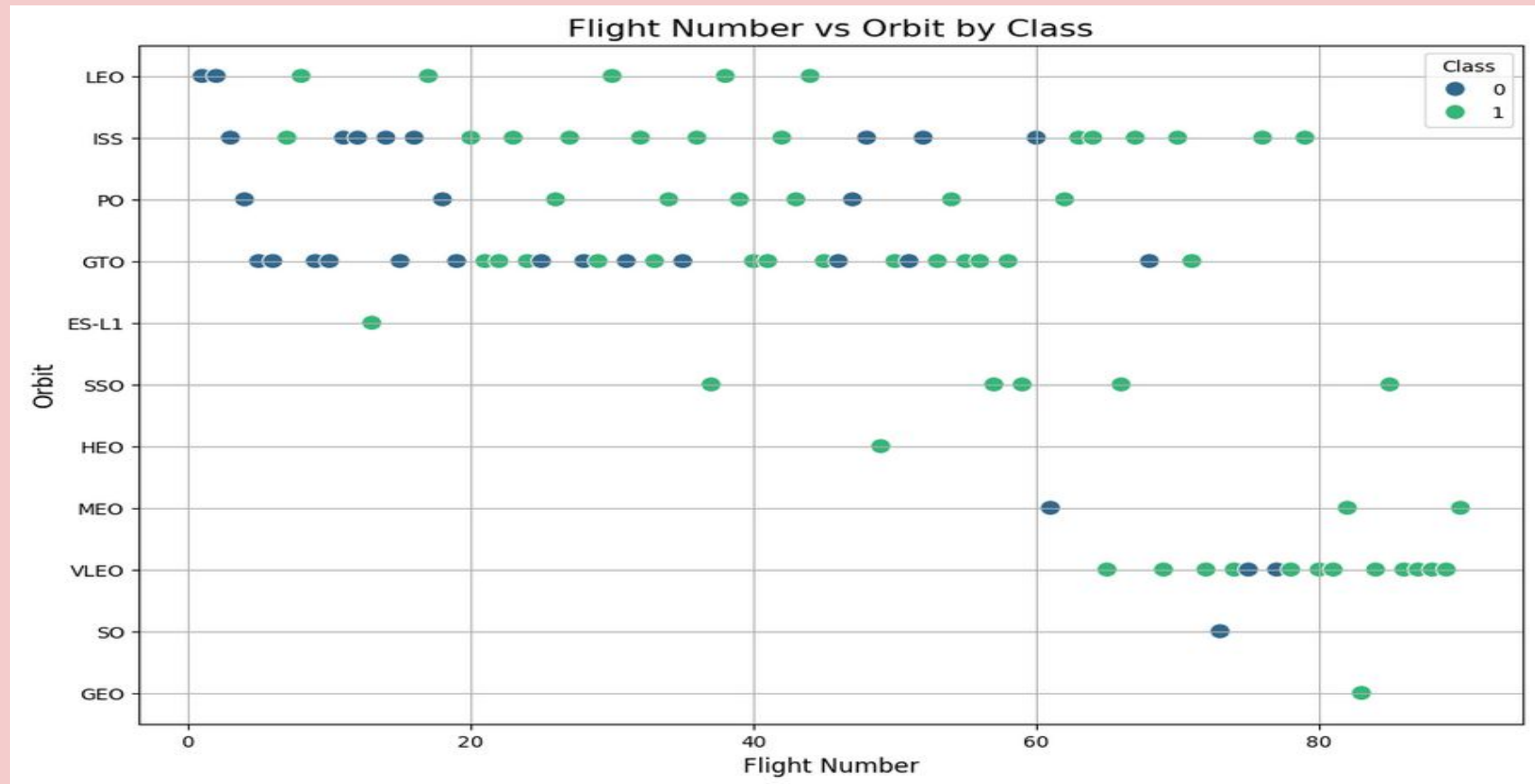
- ES-L1 (1), GEO(1), HEO(1) = have 100% success rate
- SSO (5) = has 100% success rate
- VLEO (14) = decent success rate and attempts
- SO (1) = 0% success rate
- GTO (27) = has around 50% success rate but largest sample

\* () Sample Sizes in Parenthesis





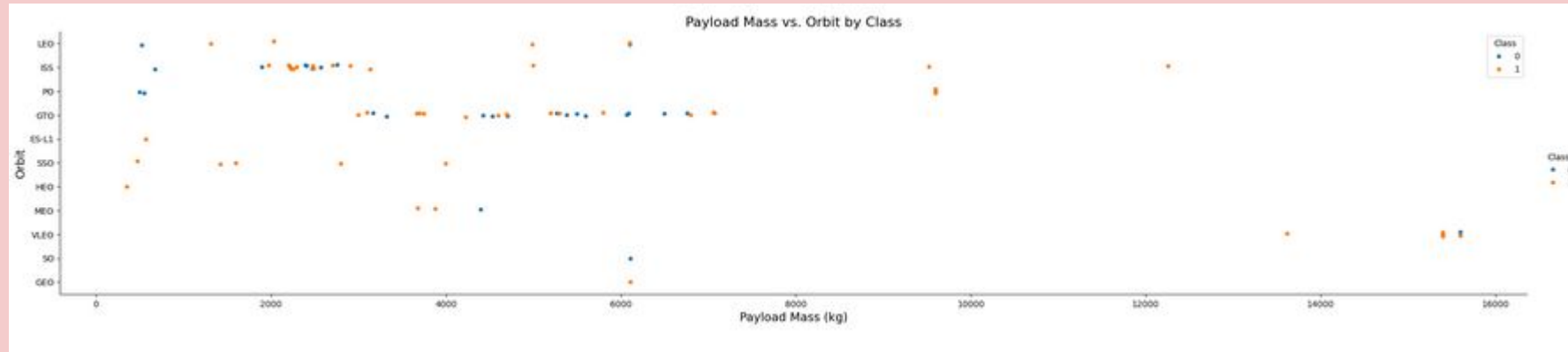
# Flight Number vs. Orbit Type



Green = Successful  
Blue = Unsuccessful

Launch Orbit preferences changed over flight number. Launch Outcome seems to correlate with this preference. SpaceX started with LEO orbits which saw moderate success and returned to VLEO in recent launches. SpaceX looks to perform better in lower orbits or Sun-synchronous orbits.

# Payload vs. Orbit Type



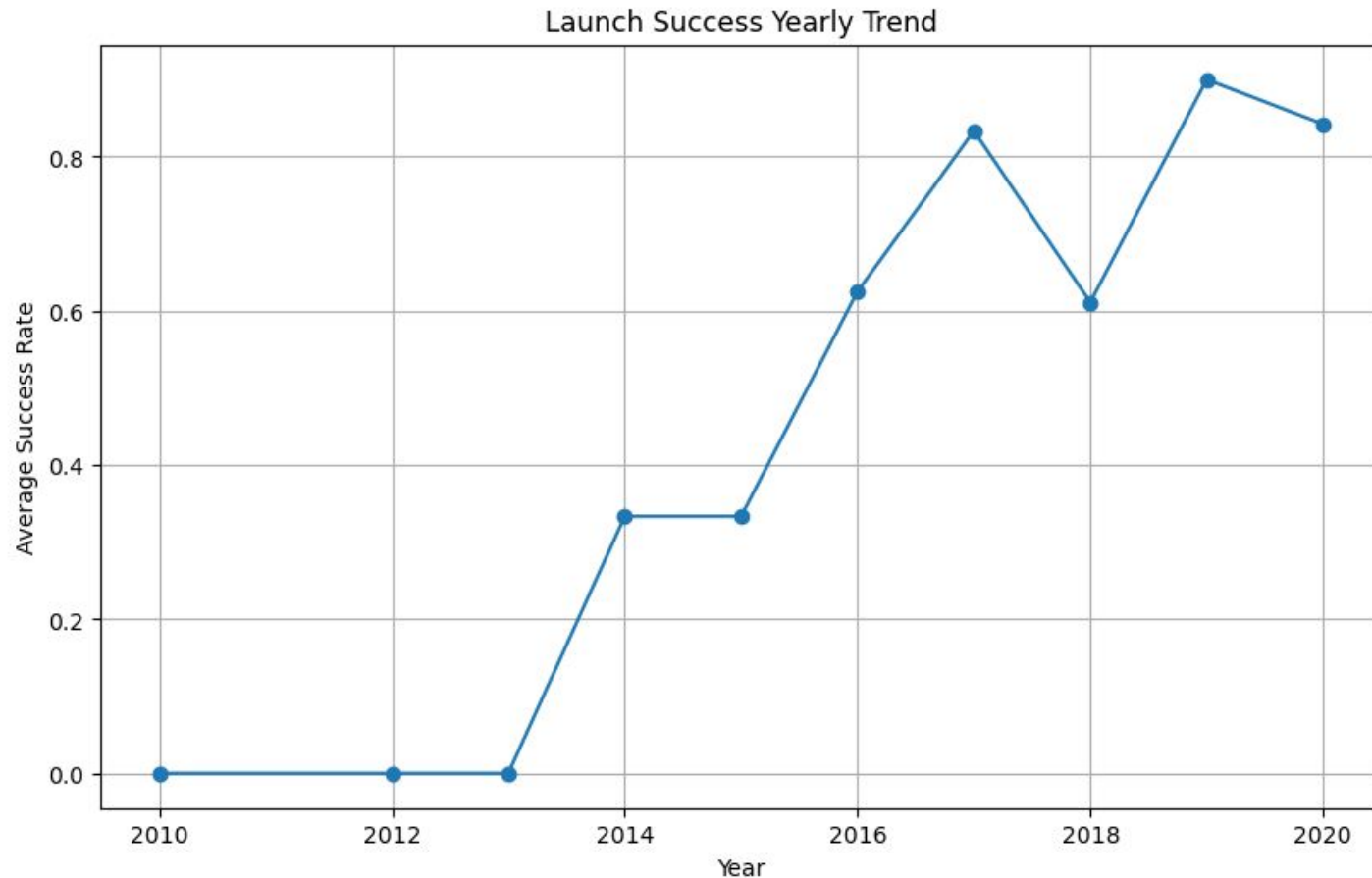
Orange = Successful

Blue = Unsuccessful

Payload mass correlates with orbit, as LEO and SSO have relatively low payload mass.

The other most successful orbit is VLEO which only has payload mass values in the higher end of the range.

# Launch Success Yearly Trend



- Success increases gradually in 2013 with a slight dip in 2018.
- In recent years success is at around 80%

95% confidence interval

# All Launch Site Names

Query for unique launch site names...

```
[10]: %%sql
      SELECT DISTINCT launch_site
      FROM SPACEXTABLE;

* sqlite:///my_data1.db
Done.

[10]: Launch_Site
      CCAFS LC-40
      VAFB SLC-4E
      KSC LC-39A
      CCAFS SLC-40
```

- CCAFS SLC-40 and CCAFSSLC-40 both represent the same launch site with data entry errors.
- CCAFS LC-40 was the previous name
- CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E only unique launch\_site values



# Launch Site Names Begin with 'CCA'

```
%%sql
SELECT *
FROM SPACEXTABLE
WHERE launch_site LIKE 'CCA%'
LIMIT 5;
```

```
* sqlite:///my_data1.db
Done.
```

| Date       | Time (UTC) | Booster_Version | Launch_Site | Payload                                                       | PAYLOAD_MASS_KG_ | Orbit     | Customer        | Mission_Outcome | Landing_Outcome     |
|------------|------------|-----------------|-------------|---------------------------------------------------------------|------------------|-----------|-----------------|-----------------|---------------------|
| 2010-06-04 | 18:45:00   | F9 v1.0 B0003   | CCAFS LC-40 | Dragon Spacecraft Qualification Unit                          | 0                | LEO       | SpaceX          | Success         | Failure (parachute) |
| 2010-12-08 | 15:43:00   | F9 v1.0 B0004   | CCAFS LC-40 | Dragon demo flight C1, two CubeSats, barrel of Brouere cheese | 0                | LEO (ISS) | NASA (COTS) NRO | Success         | Failure (parachute) |
| 2012-05-22 | 7:44:00    | F9 v1.0 B0005   | CCAFS LC-40 | Dragon demo flight C2                                         | 525              | LEO (ISS) | NASA (COTS)     | Success         | No attempt          |
| 2012-10-08 | 0:35:00    | F9 v1.0 B0006   | CCAFS LC-40 | SpaceX CRS-1                                                  | 500              | LEO (ISS) | NASA (CRS)      | Success         | No attempt          |
| 2013-03-01 | 15:10:00   | F9 v1.0 B0007   | CCAFS LC-40 | SpaceX CRS-2                                                  | 677              | LEO (ISS) | NASA (CRS)      | Success         | No attempt          |

First 5 entries in the database with the Launch Site name beginning with 'CCA'

# Total Payload Mass

```
[2]: %%sql
SELECT SUM(payload_mass__kg_) AS total_payload_mass
FROM SPACEXTABLE
WHERE customer = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db
Done.
```

```
[2]: total_payload_mass
```

| total_payload_mass |
|--------------------|
| 45596              |

- This query sums the total payload mass in kg where NASA was the customer.
- CRS stands for Commercial Resupply Services which indicates that these payloads were sent to the International Space Station (ISS).

# Average Payload Mass by F9 v1.1

```
] : %%sql
SELECT AVG(PAYLOAD_MASS_KG_) AS average_payload_mass
FROM SPACEXTABLE
WHERE booster_version = 'F9 v1.1';

* sqlite:///my_data1.db
Done.

] : average_payload_mass
_____
2928.4
```

- This query calculates the average payload mass of launches which used **Booster Version F9 v1.1**
- Average Payload Mass of the F9 v1.1 is on the low end of the payload mass range.

# First Successful Ground Landing Date

```
[14]: %%sql
SELECT MIN(Date) AS First_Successful_Landing_Date
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success' AND Landing_Outcome = 'ground pad';

* sqlite:///my_data1.db
Done.
[14]: First_Successful_Landing_Date

2015-12-22
```

- This query returns the first successful ground pad landing date.
- First ground pad landing wasn't until the end of 2015.
- Successful landings generally start to appear starting in 2014.

# Successful Drone Ship Landing with Payload between 4000 and 6000

```
%%sql
SELECT DISTINCT Booster_Version
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (drone ship)'
AND PAYLOAD_MASS__KG_ > 4000
AND PAYLOAD_MASS__KG_ < 6000;
```

```
* sqlite:///my_data1.db
```

Done.

**Booster\_Version**

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

This query returns the four [Booster Versions](#) that had successful drone ship landings and a payload mass between 4000kg and 6000kg noninclusively.



# Total Number of Successful and Failure Mission Outcomes

```
%%sql
SELECT "Mission_Outcome", COUNT(*) AS total_count
FROM SPACEXTABLE
GROUP BY "Mission_Outcome";
```

\* sqlite:///my\_data1.db

Done.

| Mission_Outcome                  | total_count |
|----------------------------------|-------------|
| Failure (in flight)              | 1           |
| Success                          | 98          |
| Success                          | 1           |
| Success (payload status unclear) | 1           |

- This query returns a count of each mission outcome.
- SpaceX appears to achieve its mission outcome nearly 99% of the time.
- This means that most of the landing failures are intended.
- 1 launch has an unclear payload status
- 1 launch failed in flight

# Boosters Carried Maximum Payload

```
%%sql
SELECT booster_version
FROM SPACEXTABLE
WHERE PAYLOAD_MASS_KG_ = (
    SELECT MAX(PAYLOAD_MASS_KG_)
    FROM SPACEXTABLE
);
```

```
* sqlite:///my_data1.db
Done.
```

## Booster\_Version

|               |
|---------------|
| F9 B5 B1048.4 |
| F9 B5 B1049.4 |
| F9 B5 B1051.3 |
| F9 B5 B1056.4 |
| F9 B5 B1048.5 |
| F9 B5 B1051.4 |
| F9 B5 B1049.5 |
| F9 B5 B1060.2 |
| F9 B5 B1058.3 |
| F9 B5 B1051.6 |
| F9 B5 B1060.3 |
| F9 B5 B1049.7 |

- This query returns the **booster versions** that carried the highest payload mass of **15600kg**.

These booster versions are very similar and all are of the **F9 B5 B10xx.x** variety, which indicates payload mass correlates with the booster version that is used.



# 2015 Launch Records

```
%%sql
SELECT
  substr(Date, 6, 2) AS Month,
  "Landing_Outcome",
  Booster_Version,
  Launch_Site
FROM
  SPACEXTABLE
WHERE
  substr(Date, 0, 5) = '2015'
  AND "Landing_Outcome" = 'Failure (drone ship)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

| Month | Landing_Outcome      | Booster_Version | Launch_Site |
|-------|----------------------|-----------------|-------------|
| 01    | Failure (drone ship) | F9 v1.1 B1012   | CCAFS LC-40 |
| 04    | Failure (drone ship) | F9 v1.1 B1015   | CCAFS LC-40 |

This query returns a list of failed landings in the year of 2015, it highlights the [Booster Version](#), Launch Site, and landing outcome.

# Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%%sql
SELECT
    "Landing_Outcome",
    COUNT(*) AS Outcome_Count
FROM
    SPACEXTABLE
WHERE
    Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY
    "Landing_Outcome"
ORDER BY
    Outcome_Count DESC;
```

```
* sqlite:///my_data1.db
Done.
```

| Landing_Outcome        | Outcome_Count |
|------------------------|---------------|
| No attempt             | 10            |
| Success (drone ship)   | 5             |
| Failure (drone ship)   | 5             |
| Success (ground pad)   | 3             |
| Controlled (ocean)     | 3             |
| Uncontrolled (ocean)   | 2             |
| Failure (parachute)    | 2             |
| Precluded (drone ship) | 1             |

- This query returns a list of both successful and unsuccessful landings with the various landing methods.
- There are 3 types of landing shown, Drone Ship, Ground Pad, and Ocean.
- Drone ship has the highest success, but equal in failures with one precluded.
- Oddly ground pad shows only 3 success but no failures

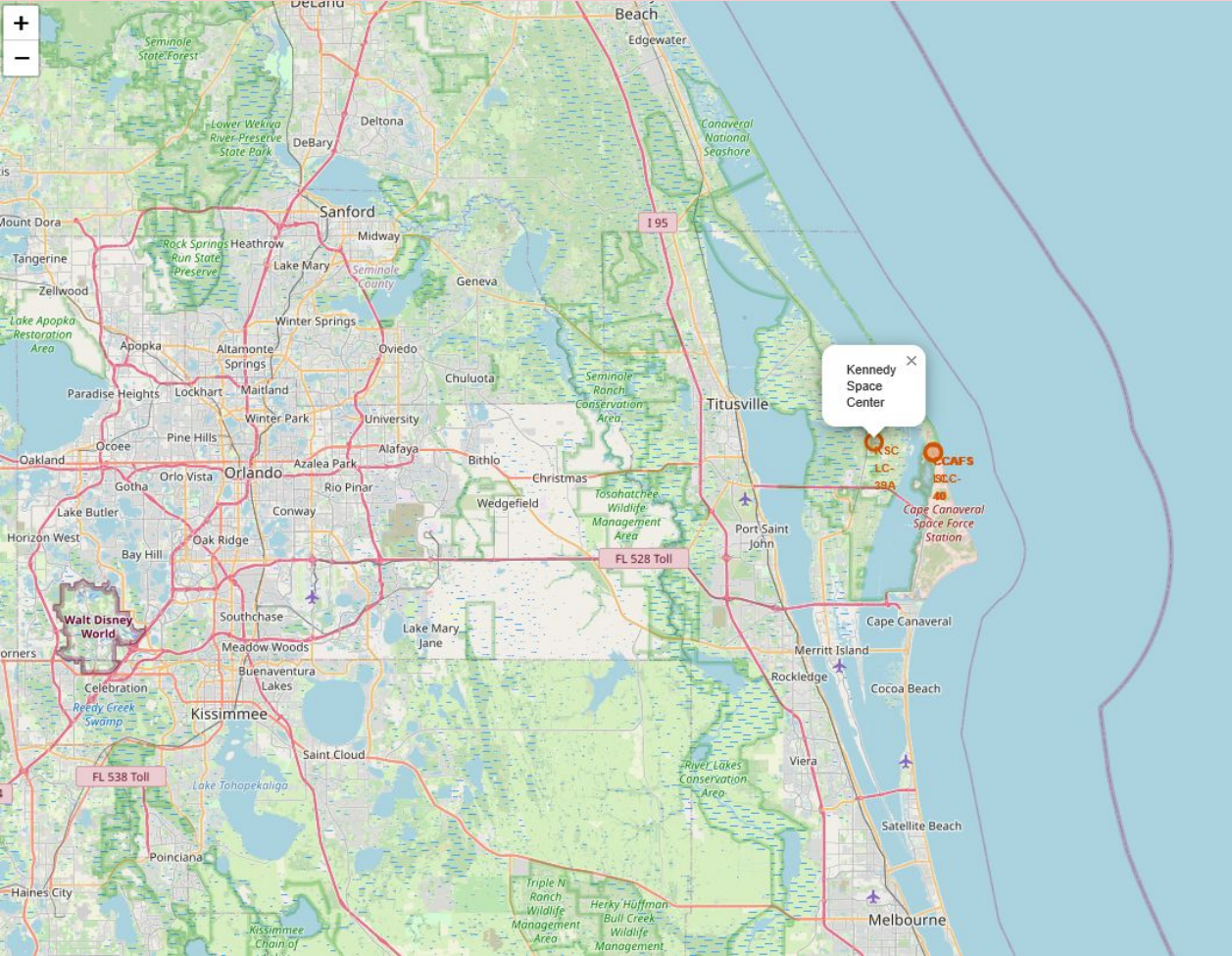
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky with stars and a view of the Earth's surface from space. The Earth's surface is mostly dark, with a dense network of yellow and orange lights representing city lights at night. The lights are concentrated in the lower right portion of the image, following the curve of the Earth. The upper portion of the image shows the dark blue sky with a few stars.

Section 3

# Launch Sites Proximities Analysis



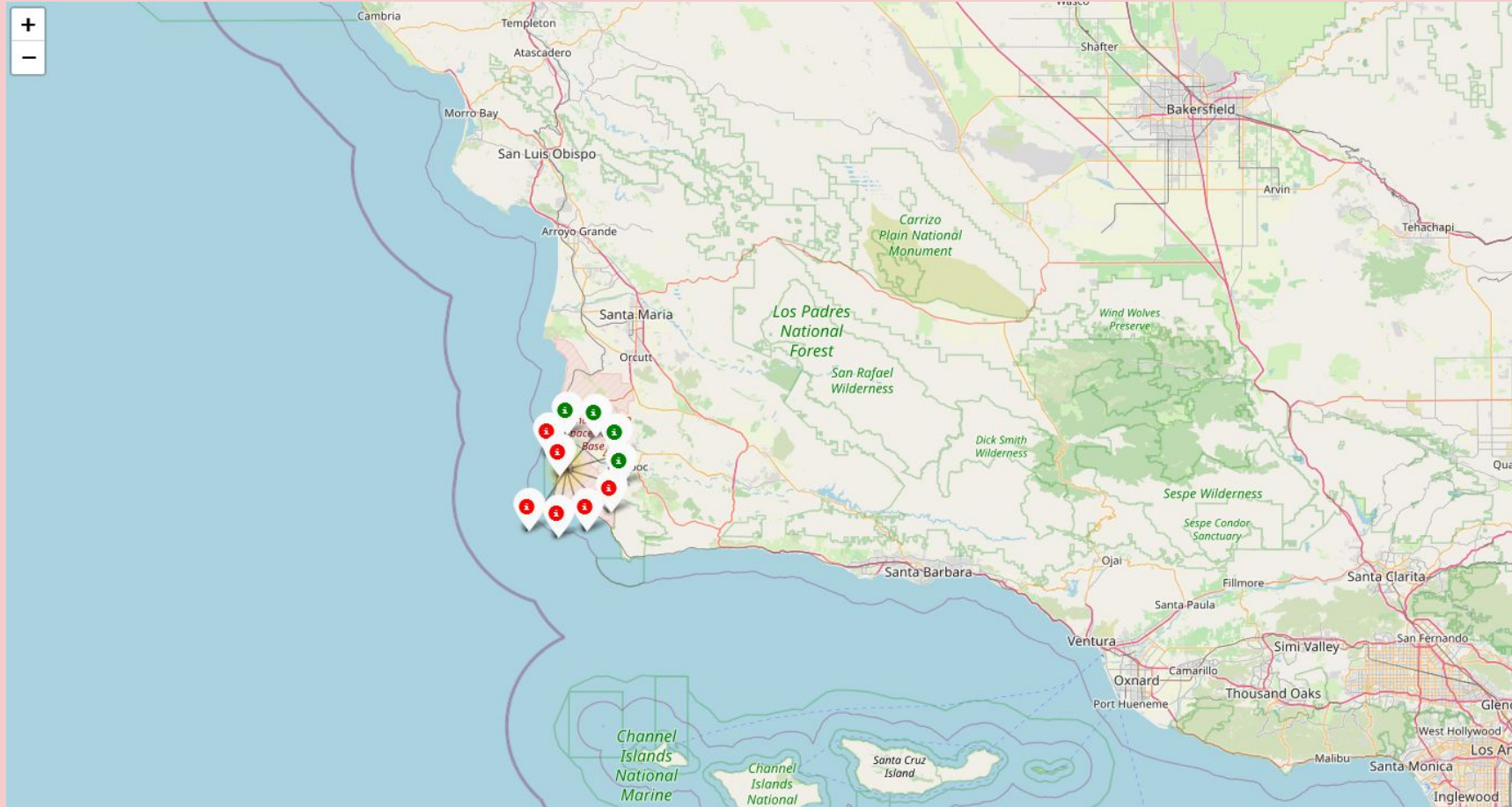
A detailed map of Southern California, highlighting the Vandenberg Space Force Base (VAFB) and its surroundings. The map includes major cities like San Luis Obispo, Santa Barbara, Los Angeles, and San Diego. Key geographical features such as the Channel Islands National Marine Sanctuary, Los Padres National Forest, and various wilderness areas are labeled. The Vandenberg Space Force Base is marked with a red dot and a callout box. The map also shows the Pacific Ocean to the west and south, and the Colorado Desert to the east.



- The left map shows the launch site on the west coast of the US map.
- The right map shows 2 launch sites on the east coast in Florida.

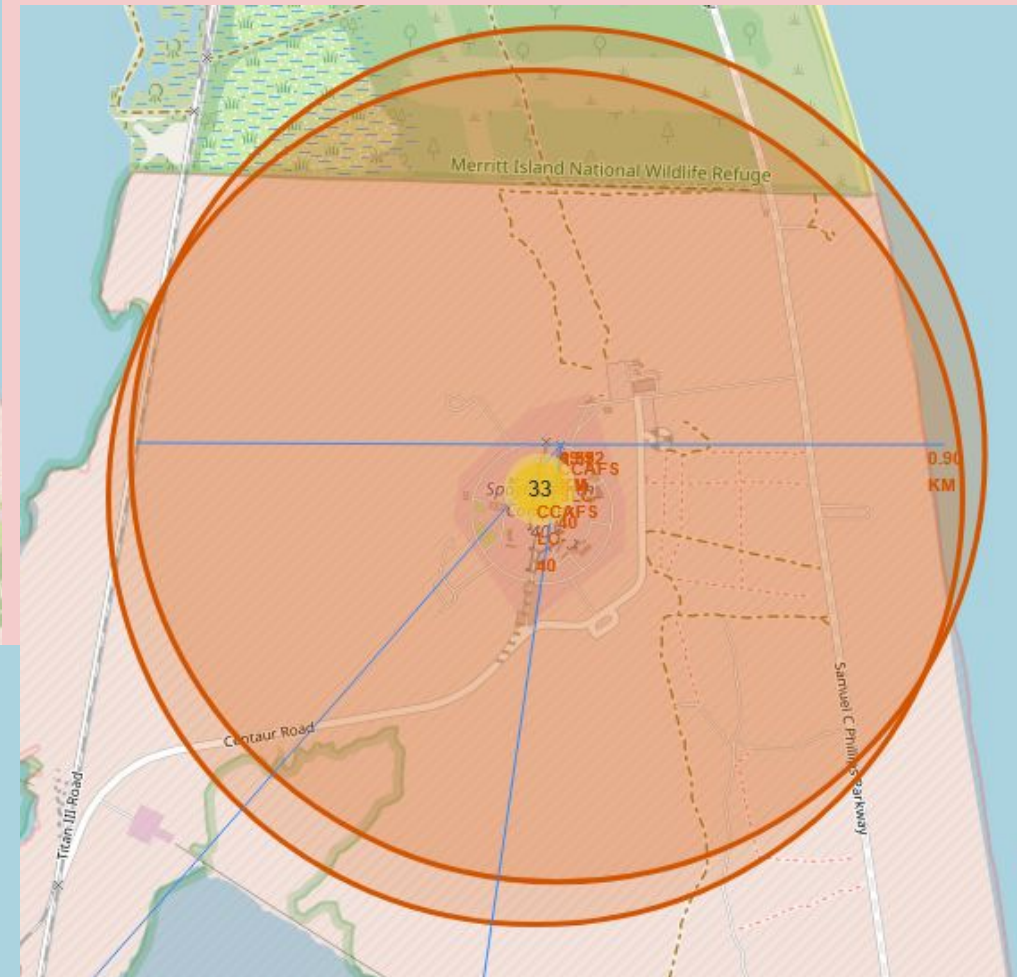
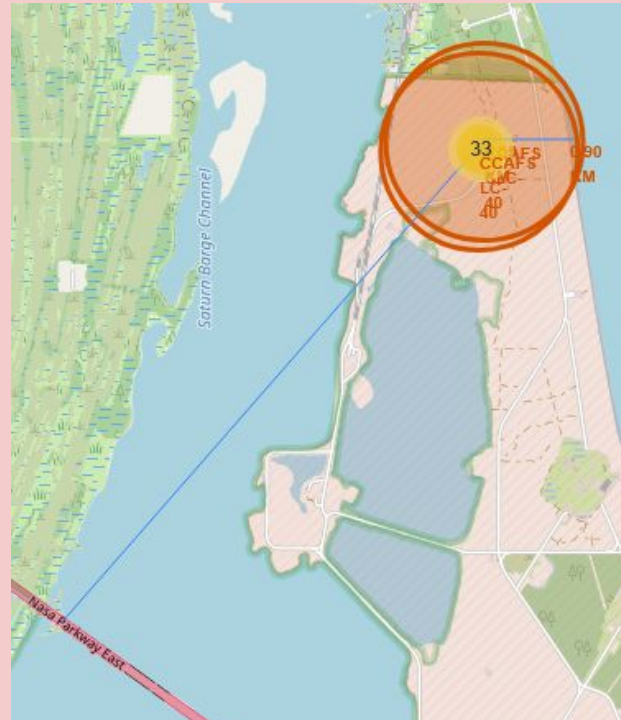
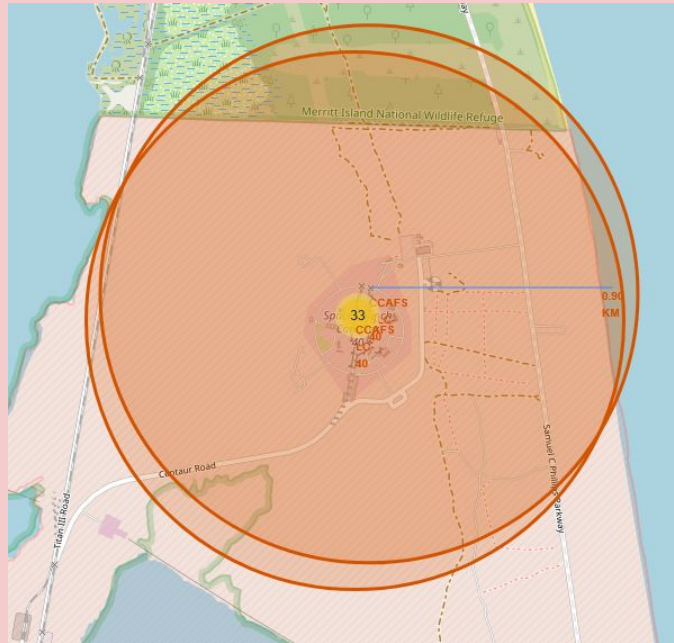


# Color-Coded Launch Markers



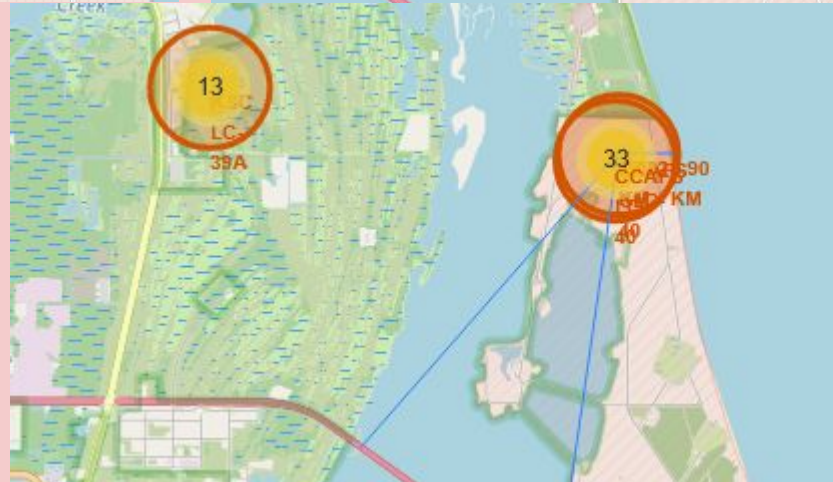
Clusters on the map can be clicked on to show each successful landing, green icon, and failed landing, red icon. In this image VAFB SLC-4E shows 4 successful landings and 6 failed landings.

# Key Location Proximities

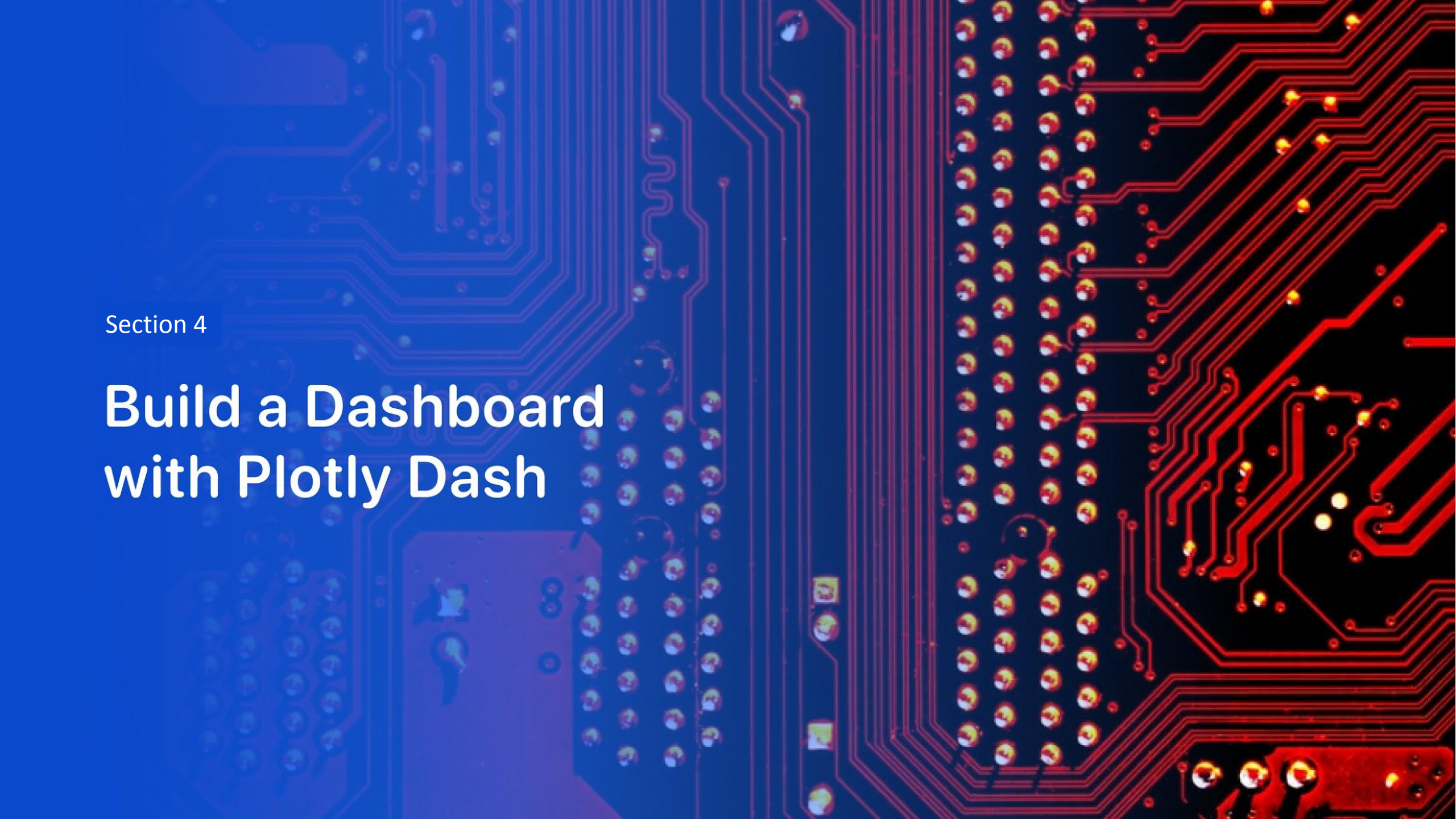


Using KSC LC-39A as an example, launch sites are very close to railways for part and supply transportation. Launch sites are close to highways for human and supply transport.

Sites are located far from large cities and close to oceans so launch failures can land in the ocean and any debris/ falling rocket will avoid densely populated areas.



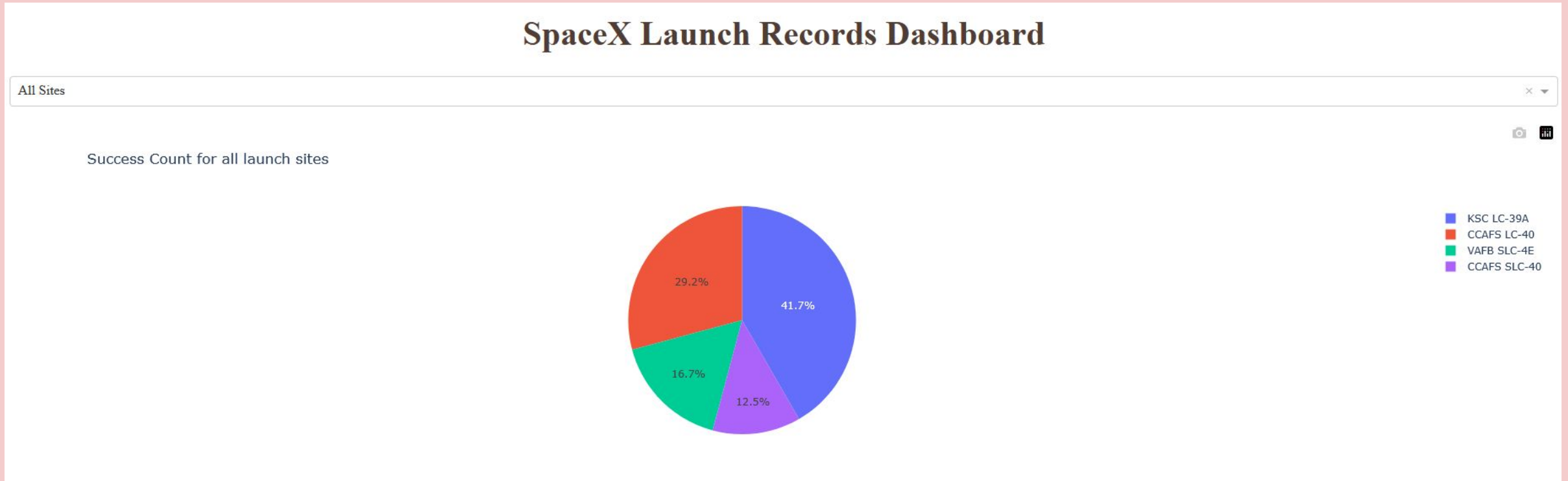




Section 4

# Build a Dashboard with Plotly Dash

# Successful Launches for Each Launch Site



The [Pie Chart](#) above shows the success rate of landings for each launch site. CCAFS LC-40 is the old name of CCAFS SLC-40, so CCAFS and KSC have the same amount of successful landings, but a majority of the successful landings were performed before the name change. VAFB has the smallest share of successful landings, possibly due to the smaller sample and increased difficulty of launching on the west coast.

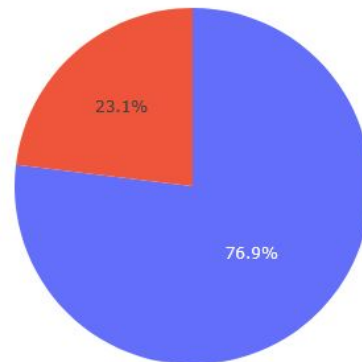
# Highest Success Rate of Launch Site

## SpaceX Launch Records Dashboard

KSC LC-39A



Total Success Launches for site KSC LC-39A

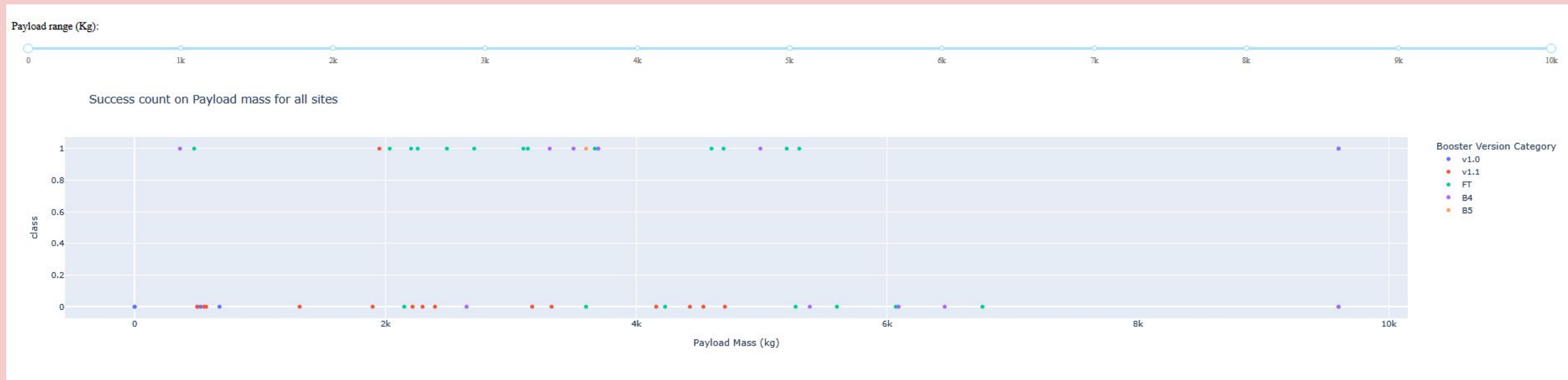


Blue = Success

KSC LC 39A has the highest success rate with 10 successful landings and 3 failed landings.



# Payload Mass vs Success vs Booster Version Category

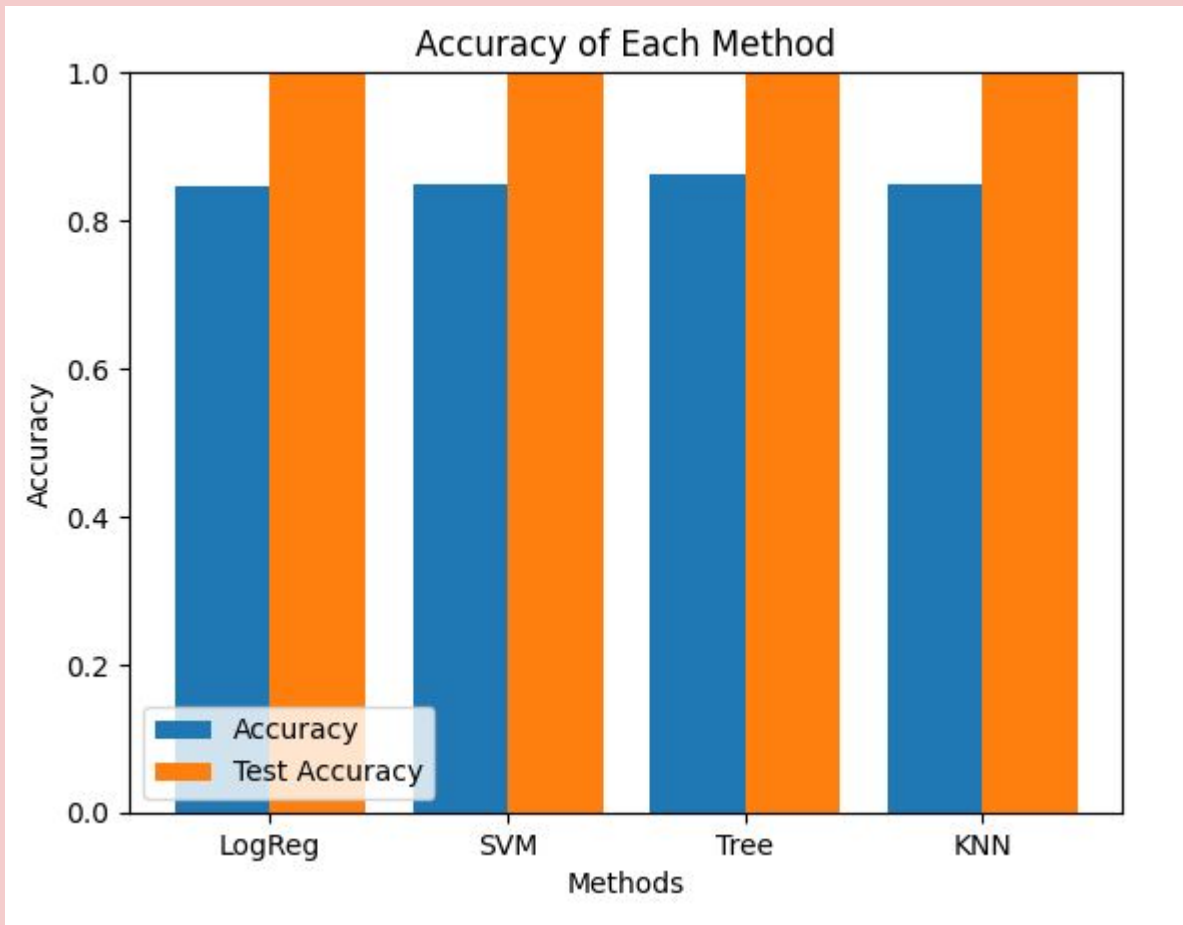


The plotly dashboard has a Payload range selector, this allows you to view the success vs fail with specific payloads. This scale is set from 0 to 10000Kg instead of the max payload of 15600Kg. Class 1 indicates a success or True landing, while 0 indicates a failure or False landing. Scatter plot represents the different booster versions by color featured in the legend on the right of the graph. A majority of successful landings happen within a payload range of 2000Kg and 4000Kg.

Section 5

# Predictive Analysis (Classification)

# Classification Accuracy



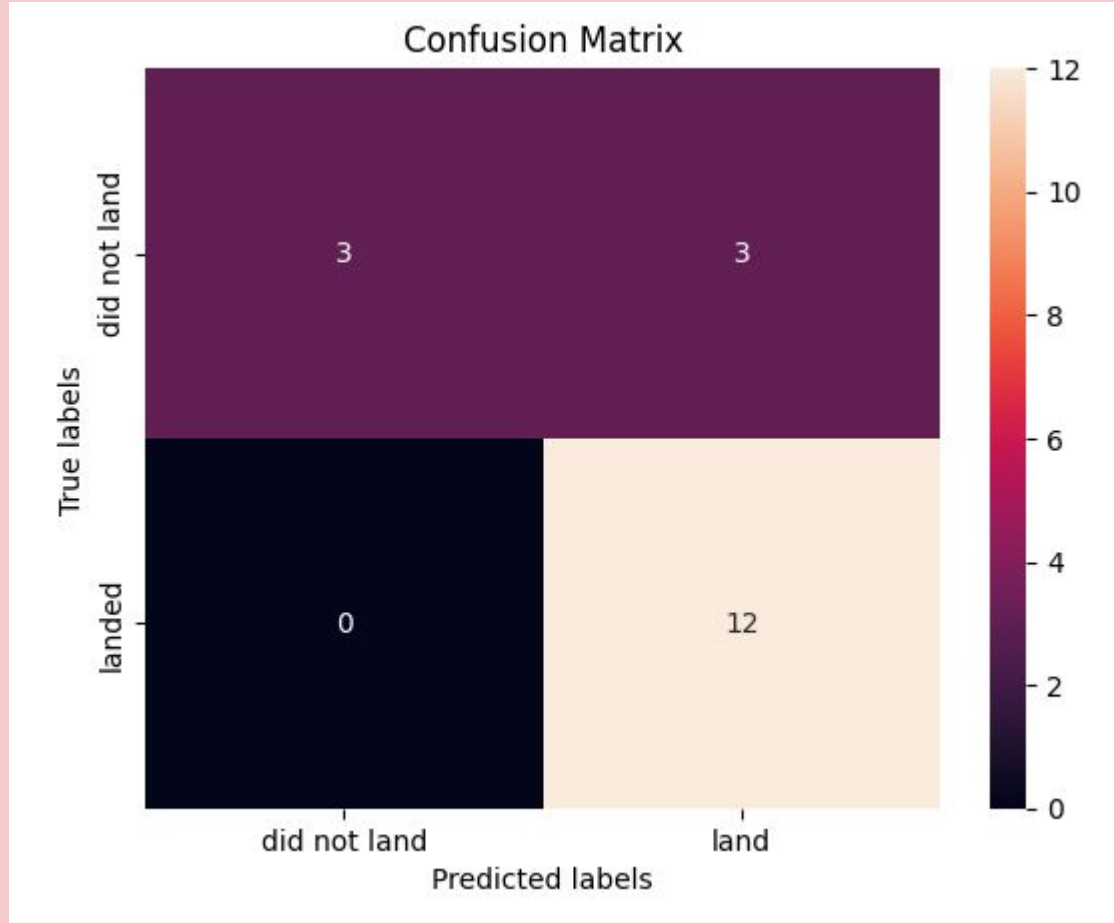
All models had virtually the same accuracy on the test set at 83.33% accuracy rate.

One must consider that the test size is fairly small coming in at a sample size of only 18.

Due to this small test size it can cause large variance in accuracy results, such as those in the Decision Tree Classifier model in repeated runs.

It would be wise to acquire more data to determine the best model.

# Confusion Matrix



Due to all models performing the same for the test set, the confusion matrix is the same across all models.

The models predicted 12 successful landings when the True label was a success, predicted 3 unsuccessful landings when the True label was unsuccessful, predicted 3 successful landings when the True label was unsuccessful (False Positive).

The models over predict successful landings.

# Conclusions

- Our Task: to develop a machine learning model for Space Y who wants to bid against SpaceX
- The goal of the model is to predict when Stage 1 will successfully land, saving \$100 million USD
- Used data from a public SpaceX API and web scraping SpaceX wikipedia page
- Created data labels and stored data into a DB2 SQL database
- Created a dashboard for visualization
- Created a machine learning model with an accuracy of 83%
- Space Y can use this model to predict with relatively high accuracy whether a launch will have a successful Stage 1 landing before launch to determine whether the launch should be made or not
- If possible more data should be collected to better determine the best machine learning model and improve accuracy



# Appendix

Github repository url:

<https://github.com/Blasko1369/SpaceY>

Instructors:

Rav Ahuja, Alex Aklson, Aije Egwaikhide, Svetlana Levitan, Romeo Kienzier, Polong Lin, Joseph Santarcangelo, Azim Hirjani, Hima Vasudevan, Saeed Aghabozorgi, Yan Luo

**A Big Thank you to all the instructors and fellow peers!**

Thank you!

